

0.1 Minimal Polynomial

Definition 0.1.1. Let V be a Vector Space over F , and let $T : V \rightarrow V$ be a linear operator. A map

$$\varphi_T : F[x] \rightarrow \mathcal{L}(V) : p(x) \mapsto p(T)$$

is called *evaluation* of T . It is a ring-homomorphism, and $\ker \varphi_T$ is principal ideal. Moreover,

$$F[x] / \ker \varphi_T \cong F[T]$$

by the first-isomorphism theorem.

Note: $\mathcal{L}(V) = \{T : V \rightarrow V \mid T \text{ is linear operator}\}$ has algebraic structures:

$(\mathcal{L}(V), +, \cdot)$ is Vector Space over F and $(\mathcal{L}(V), +, \circ)$ is a Ring with identity.

Generally, $(\mathcal{L}(V), +, \circ)$ is non-commutative, and Not domain. For instance, let $V = F^n$. Then, for each $1 \leq j \leq n$,

$$A_j : F^n \rightarrow F^n : (a_i)_{i=1}^n \mapsto (0, \dots, a_j, 0, \dots)$$

are not zero maps, but $E_1 \circ E_2 = \mathbf{0}$. And ask a toddler on the street, examples for Non-Commutative.

But, the subring

$$F[T] = \{p(T) \mid p(x) \in F[x]\} \subset \mathcal{L}(V)$$

is Commutative Ring with identity, still it is not a domain, in general. Moreover, as a Subspace,

$$F[T] = \text{span}_F \{T^n \mid n = 0, 1, \dots\}$$

If V is finite-dimensional space, then $\mathcal{L}(V)$ has a n^2 dimension, therefore

$$T^0, T^1, \dots, T^{n^2}$$

is linearly dependent, this implies there exists a polynomial $p(x) \in F[x]$ such that $p(T) = 0$. In other words,

If V has a finite dimension, then for linear operator $T : V \rightarrow V$, $\ker \varphi_T$ is non-zero principal ideal.

In this case, the unique monic polynomial of $\ker \varphi_T$ is called the *minimal polynomial* for T .

Thm. The characteristic and minimal polynomials for T have the same roots.

Proof. Let p be the minimal polynomial for T .

Suppose that $c \in F$ is root of p . That is, $p(c) = 0$. Then,

$$p = (x - c) \cdot q \quad (\deg q < \deg p)$$

for some $q \in F[x]$. Since p is the minimal, $q(T) \neq 0$. Take $\beta \in V$ s.t. $q(T)\beta \neq 0$. Then, $0 = p(T)(\beta) = (T - cI) \cdot q(T)(\beta)$. Thus $T \cdot q(T)(\beta) = c\beta$, c is eigenvalue.

Conversely, let $c \in F$ be an eigenvalue of T . That is, $T\beta = c\beta$ for some nonzero $\beta \in V$. Now, $0 = p(T)(\beta) = p(c)\beta$ and since β is nonzero, $p(c) = 0$. ~~QED~~

Conductor.

Def. Let W be an invariant subspace for $T: V \rightarrow V$, and $\alpha \in V$ be a vector. Define T -Conductor of α into W as;

$$S_T(\alpha; W) \stackrel{\text{def}}{=} \{g \in F[x] \mid g(T)(\alpha) \in W\}.$$

In particular, if $W = \{0\}$, the Conductor is called the T -annihilator of α .

Lemma. The Conductor $S(\alpha; W)$ is an Ideal in $F[x]$.

Proof. First, if $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, because for any $w \in W$,
 $\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(T(\beta)) = \dots = T(T^{n-1}(\beta)) \in W$.

And since $W \subseteq V$, it is closed under the addition and scalar multiplication,
 $g(T)[W] \subseteq W$. Now, to show $S(\alpha; W)$ is an Ideal, first show that it is space.

For any $f, g \in S(\alpha; W)$ and $c \in F$, $(cf + g)(T) = cf(T) + g(T)$, therefore

$$(cf + g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is subspace. Moreover, for any $f \in F[x]$ and any $g \in S(\alpha; W)$,

$$(fg)(T)(\alpha) = f(T) \underbrace{g(T)(\alpha)}_{\in W} \in W.$$

Thus $fg \in S(\alpha; W)$. $\square$

Note: Given the linear operator T and T -invariant subspace W , and any vector $\alpha \in V$,

$$Q: F[x] \rightarrow V/W: f \mapsto f(T)(\alpha) + W$$

is $F[x]$ -module Homomorphism, so $S_T(\alpha; W) = \text{Ker } Q$ is Principal Ideal in $F[x]$.

Lemma. Let V be a fin. dim v.s over F , and let $T: V \rightarrow V$ be a linear operator. Suppose that the minimal polynomial of T splits over F as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper T -invariant subspace $W < V$, there exists a vector $\alpha \in V$ s.t. $\alpha \notin W$ and for some c_i , $(T - c_i I)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given, and let $q \in F[x]$ be a T -annihilator of β into W . Then, q divides p . And, since $\beta \notin W$, q is not constant. Therefore,

$$q = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k} \quad (0 \leq e_i \leq r_i)$$

and at least one of e_i is positive. Let i s.t. $e_i > 0$. Then,

$$q = (x - c_i) \cdot h \quad \text{where } h = q / (x - c_i) \in F[x].$$

Since q is T -annihilator of β , $h(T)(\beta) \notin W$, but $(T - c_i I)(h(T)(\beta)) = q(T)(\beta) \in W$.

Thm. Let V be a finite dimensional vector space over F , and $T: V \rightarrow V$ be a linear. T is Triangularizable if and only if minimal polynomial of T splits over F .

Proof. If T is triangularizable with respects to a basis β , then the characteristic polynomial is

$$f(x) = \det([T]_{\beta} - xI) = \prod_{i=1}^n (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{'th diagonal element of } [T]_{\beta}$$

thus, proved. Conversely, choose $\alpha_1 \in V$ s.t. $\alpha_1 \notin \{0\}$, $(T - c_1 I)(\alpha_1) \in \{0\}$ for some eigenvalue $c_1 \in F$ of T . Choose $\alpha_2 \in V$ s.t. $\alpha_2 \notin \langle \alpha_1 \rangle$, $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, it is possible because $\langle \alpha_1 \rangle$ is T -invariant. Again, choose $\alpha_3 \in V$ s.t. $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$, $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$. It is possible again, $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$.

Now, we obtain $\{\alpha_1, \dots, \alpha_n\}$ s.t.

$$T(\alpha_1) = c_1 \alpha_1$$

$$T(\alpha_2) = a_{21} \alpha_1 + c_2 \alpha_2$$

⋮

$$T(\alpha_n) = a_{n1} \alpha_1 + a_{n2} \alpha_2 + \dots + c_n \alpha_n$$

Thm. Let V be a fin. dim. v.s over F , and $T: V \rightarrow V$ be a linear operator.

T is diagonalizable if and only if minimal polynomial of T splits and separable.

Proof. Suppose that T is diagonalizable. Let $c_1, \dots, c_k$ be distinct eigenvalues.

Then, for any eigenvector $d \in V$, there is a $c_i \in F$ s.t. $(T - c_i I)(d) = 0$.

Therefore, for $p = (x - c_1) \dots (x - c_k)$,

$$p(T) = (T - c_1 I) \dots (T - c_k I)(d) = 0$$

because $\{T - c_1 I, \dots, T - c_k I\}$ commute. Since T is diagonalizable, for any vector of V can be represented by linear combination of eigenvectors,

$$p(T) = 0$$

Thus proved. Conversely, let W be a spanned subspace by all of eigenvectors of T .

Suppose that $W \neq V$. By the construction, W is T -invariant. Therefore, by the Lemma, there is $\alpha \in V \setminus W$ and eigenvalue $c_0 \in F$ s.t. $\beta = (T - c_0 I)(\alpha) \in W$.

Since $\beta \in W$, $\beta = \beta_1 + \dots + \beta_k$ where $T(\beta_i) = c_i \beta_i$, $1 \leq i \leq k$. Therefore, for any $n \in \mathbb{Z}$

$$h(T)(\beta) = \sum_{i=1}^k h(c_i) \cdot \beta_i \in W.$$

Put $p = (x - c_0)q$. Also, $q - q(c_0) = (x - c_0)h$. Then,

$$q(T)(\alpha) - q(c_0)(\alpha) = h(T)(T - c_0 I)(\alpha) = h(T)(\beta) \in W$$

And, $0 = p(T)(\alpha) = (T - c_0 I)q(T)(\alpha) = q(T)(T - c_0 I)(\alpha) \in W$.

Therefore $q(c_0)(\alpha) \in W$, and since $\alpha \notin W$, $q(c_0) = 0$. Contradicts to $p = (x - c_0)^2 q$.

Summary.

Let $T: V \rightarrow V$ be a linear operator on the fin. dim. V over F .

For the characteristic polynomial $f(t) \in F[t]$, and the minimal polynomial $p(t) \in F[t]$, $p(t)$ divides $f(t)$ by the Cayley-Hamilton Thm, and for any T -invariant $W \leq V$ and $\alpha \in V$, every T -annihilator of α into W divides $p(t)$. Moreover, $p(t)$ and $f(t)$ share the roots, so,

T is triangularizable $\Leftrightarrow p(t)$ splits over $F \overset{\text{PIR}}{\Leftrightarrow} f(t)$ splits over $F \Leftrightarrow T$ has a Jordan form.

$p(t)$ splits over F and separable $\Leftrightarrow T$ is diagonalizable
 $\Rightarrow f(t)$ is separable.

Anyway, T has a Primary decomposition, moreover, T has a Rational-Normal form.

Calculation.

Let $D: P_n(\mathbb{R}) \rightarrow P_n(\mathbb{R}): P(x) \mapsto D_x P(x)$.

Find the minimal polynomial of D and xD , and find the Jordan form.

Proof. Clearly, for all x^m ($0 \leq m \leq n$),

$$D^{n+1}(x^m) = 0, \text{ so } D^{n+1} = 0.$$

And, $\{I, D, \dots, D^n\}$ is linearly independent, because:

Suppose that $C_0 I + C_1 D + \dots + C_n D^n = 0$.

Substituting $\frac{1}{m!} x^m$ for each $m = 0, 1, \dots, n$, then

$$0 = (C_0 I + \dots + C_n D^n) \left(\frac{1}{0!} \cdot x^0 \right) = C_0$$

$$0 = (C_0 D + \dots + C_n D^n) \left(\frac{1}{1!} \cdot x \right) = C_1$$

$\vdots$

So, $C_0 = \dots = C_n = 0$. Therefore, the minimal polynomial of D is t^{n+1} .

Meanwhile, $x D(1) = 0$,

$$x D(x) = x,$$

$$x D(x^2) = 2x^2,$$

$$\uparrow 3x^3$$

$\vdots$

Therefore, $[xD]_{\beta} = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$, where $\beta = \{1, \dots, x^n\}$ is the standard basis for $P_n(\mathbb{R})$.

So, the characteristic polynomial is $t \cdot (t-1) \cdot \dots \cdot (t-n)$, distinct $n+1$ roots, the minimal polynomial is same. Now, the Jordan-normal form are

$$[D] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & \ddots & 0 \end{bmatrix}, \quad [xD] = \begin{bmatrix} 0 & 0 & 0 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \text{ (diagonal)}.$$

Jordan and Rational form of 3×3 Matrix.

$$A = \begin{pmatrix} 0 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix}$$

$$t^3 - 4t^2 + 4$$

$$\begin{aligned} f(t) = \det(-A + tI) &= \begin{vmatrix} t & 3 & -7 \\ 1 & t+1 & -5 \\ 1 & 2 & t-6 \end{vmatrix} = t \cdot [(t+1)(t-6) + 10] - (3t-18+14) + (-15+7t+7) \\ &= t(t^2-5t+4) - (3t-4) + (7t-8) \\ &= t^3 - 5t^2 + 4t - 3t + 4 + 7t - 8 \\ &= t^3 - 5t^2 + 8t - 4 \\ &= (t-1)(t-2)^2. \end{aligned}$$

Clearly, $A - I \neq 0$ and $A - 2I \neq 0$.

By the Cayley-Hamilton, $f(A) = 0$. So, we will check that either $(A - I)(A - 2I)$ or $(A - 2I)^2$ is zero. $\leftarrow$ has same roots, so we necessary.

$$1). (A - I)(A - 2I) = \begin{pmatrix} -1 & -3 & 7 \\ -1 & -2 & 5 \\ -1 & -2 & 5 \end{pmatrix} \cdot \begin{pmatrix} -2 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{pmatrix} = \begin{pmatrix} 2+3-7 & \dots & \dots \end{pmatrix} \neq 0.$$

So, $f(t)$ is the characteristic polynomial and the minimal polynomial.

Now, $f(t)$ is not separable but splits, so A has the Jordan form

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

And has a rational form

$$\begin{bmatrix} 0 & 0 & 4 \\ 1 & 0 & -8 \\ 0 & 1 & 5 \end{bmatrix}$$

Rational form Exercise.

$$\text{Let } A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}$$

$$\begin{aligned} \det(tI - A) &= \begin{vmatrix} t-5 & 6 & 6 \\ 1 & t-4 & -2 \\ -3 & 6 & t+4 \end{vmatrix} = (t-5)(t^2-6t+12) - (6t+24-36) - 3(-12-6t+24) \\ &= (t-5)(t^2-4) - (6t-12) + 3(6t-12) \\ &= t^3-4t-5t^2+20+12t-24 \\ &= t^3-5t^2+8t-4 \\ &= (t-1)(t-2)^2. \end{aligned}$$

The possible: $(t-1)(t-2)^i$ ($i=1,2$).

$$(A-I)(A-2I) = \begin{pmatrix} 4 & -6 & -6 \\ -1 & 3 & 2 \\ 3 & -6 & -5 \end{pmatrix} \cdot \begin{pmatrix} 3 & -6 & -6 \\ -1 & 2 & 2 \\ 3 & -6 & -6 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{So, } A \sim \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \text{ where } A_1 = \begin{bmatrix} 0 & -2 \\ 1 & 3 \end{bmatrix}, A_2 = [2].$$